

Task 2: End-to-End ML Pipeline with Scikit-learn

Objective

Build a reusable, production-ready machine learning pipeline for predicting customer churn using the Telco Customer Churn dataset. The pipeline encapsulates preprocessing, model training, hyperparameter tuning, and export for deployment.

Files

File	Description
notebook.ipynb	Complete analysis with narrative and visualizations
src/data_loader.py	Data ingestion, cleaning, and feature type detection
src/pipeline_builder.py	ColumnTransformer and Pipeline construction
src/models.py	GridSearchCV training and evaluation utilities
src/visualizations.py	Reusable plotting functions
outputs/figures/	Generated plots (PNG, 200 DPI)
outputs/churn_pipeline.joblib	Exported production-ready pipeline

Quick Start

```
pip install -r ../requirements.txt
jupyter notebook notebook.ipynb
```

Key Results

- **Best Model:** Logistic Regression (Test ROC-AUC ~0.85)
- **Pipeline Architecture:** ColumnTransformer → Pipeline → GridSearchCV
- **Production Export:** Complete pipeline saved as churn_pipeline.joblib
- **Top Churn Drivers:** Tenure, Contract type, MonthlyCharges, InternetService

Tech Stack

- Python 3.10+
- pandas, numpy
- matplotlib, seaborn
- scikit-learn (Pipeline, ColumnTransformer, GridSearchCV)
- joblib (model serialization)
- Jupyter Notebook